ER to Relational Mapping

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Session Objective

- To learn ER-to-Relational Mapping

Session Outcome

At the end of this session, participants will be able to

- To Map relation from ER diagram

ER-to-Relational Mapping Algorithm

- Mapping of Regular Entity Types
- Mapping of Multivalued attributes.
- Mapping of Weak Entity Types
- Mapping of Binary 1:1 Relation Types
- Mapping of Binary 1:N Relationship Types.
- Mapping of Binary M:N Relationship Types.
- Mapping of N-ary Relationship Types.

Relational Database Design by ER-Relational Mapping

- **Relations:** (tables) Correspond with entity types and with many to many relationship type
- **Rows:** Correspond with entity instances and with many to many relationship instances
- **Columns:** Correspond with attributes

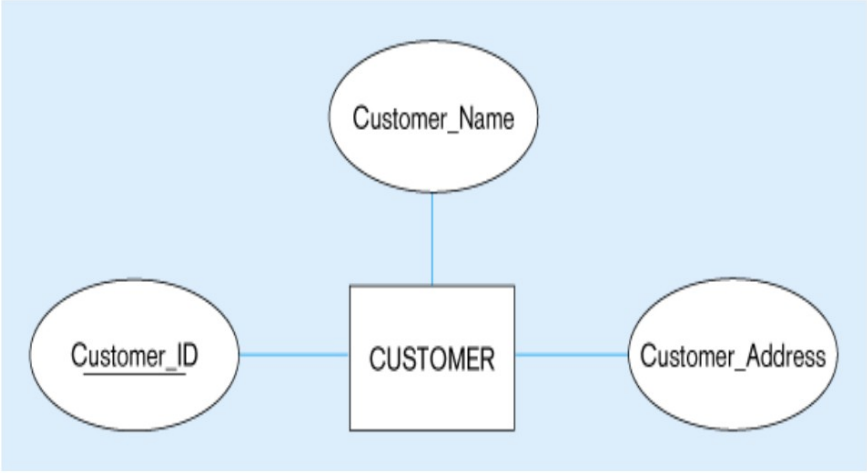
ER-to-Relational Mapping Algorithm

1. **Mapping of Regular Entity Types.**
   - For each **regular (strong) entity** type E in the ER schema, create a relation R that includes all the simple attributes of E.
   - Choose one of the **key attributes** of E as the **primary key** for R.
   - If the chosen key of **E is composite**, the **set of simple attributes** that form it will together form the **primary key** of R.

Mapping Regular Entity Types - Simple Attributes

Mapping regular entity types

(a) CUSTOMER entity type with simple attributes



(b) CUSTOMER relation

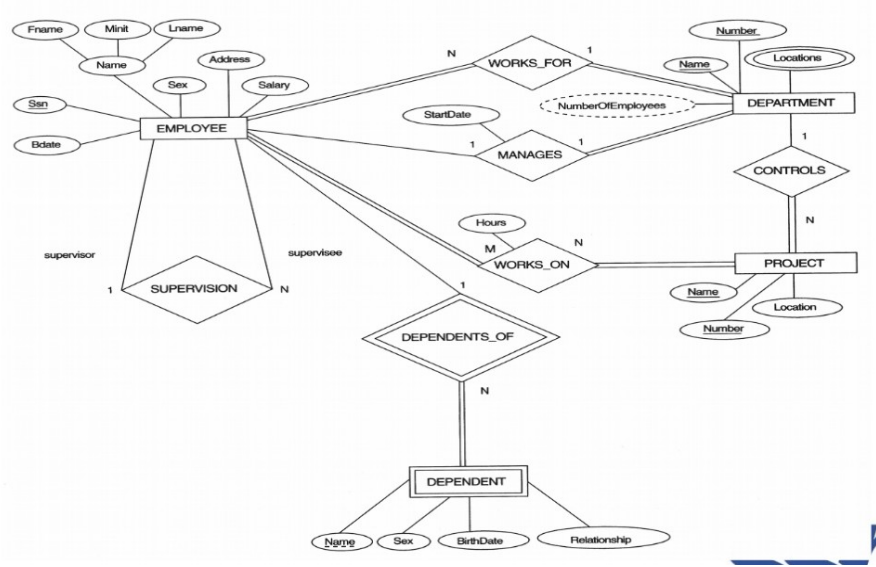
CUSTOMER

Customer_ID	Customer_Name	Customer_Address

Simple Attributes - Example

The relations **EMPLOYEE, DEPARTMENT, and PROJECT** for the COMPANY database corresponds to the regular entities in the ER diagram.

SSN, DNUMBER, and PNUMBER are the primary keys for the relations EMPLOYEE, DEPARTMENT, and PROJECT



ER to Relation

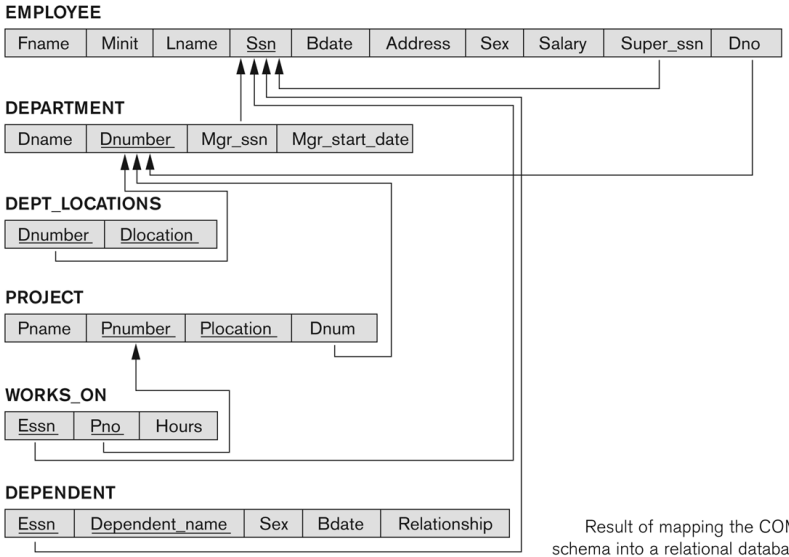


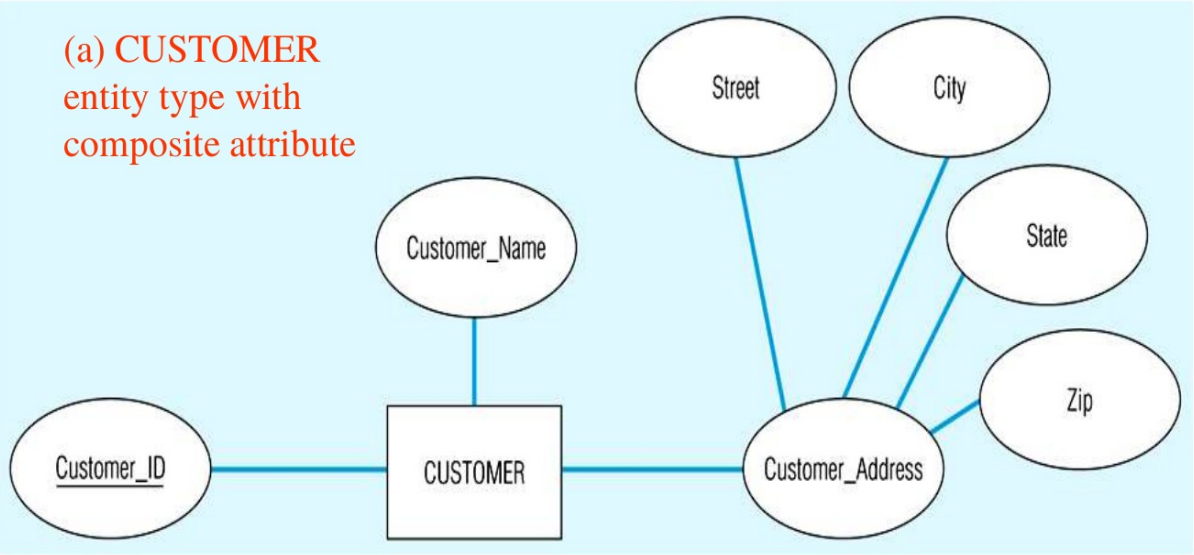
Figure 7.2
Result of mapping the COMPANY ER schema into a relational database schema.

Mapping of Different Attributes in the Relation

- **Simple attributes**: ER attributes map directly onto the relation
- **Composite attributes**: Use only their simple, component as attributes
- **Multivalued Attribute**: Becomes a separate relation with a foreign key taken from the superior entity

1. Mapping Regular Entity Types - Composite Attributes

Mapping regular entity types



(b) CUSTOMER relation with address detail

CUSTOMER

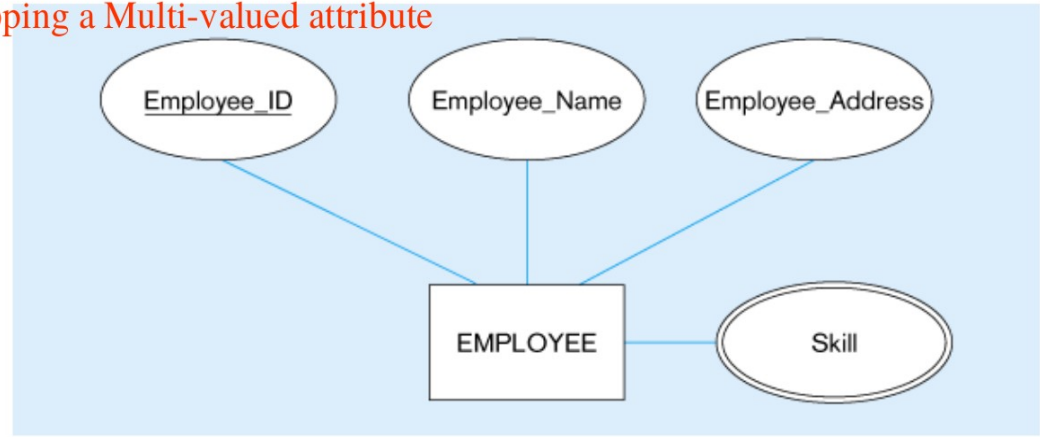
Customer_ID	Customer_Name	Street	City	State	Zip

2. Mapping Regular Entity Types - Multivalued Attributes

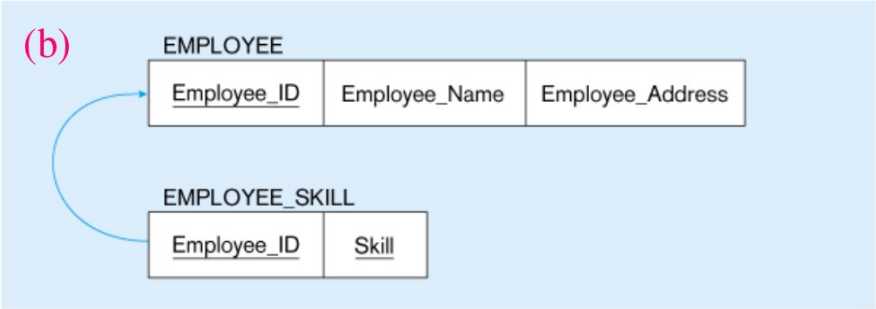
- For each multi valued attribute A, create a new relation R.
- This relation R will include:
  - The multi valued attribute **A** as its attribute.
  - The **primary key attribute K of the original relation - as a foreign key in R**
  - The primary key of R is the combination of **A and K**.
- If the multivalued attribute is composite, include it as simple components.

Mapping Regular Entity Types - Multivalued Attributes

Mapping a Multi-valued attribute



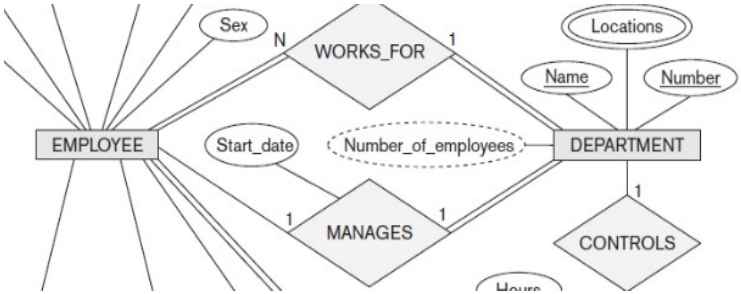
Multivalued attribute becomes a separate relation with foreign key

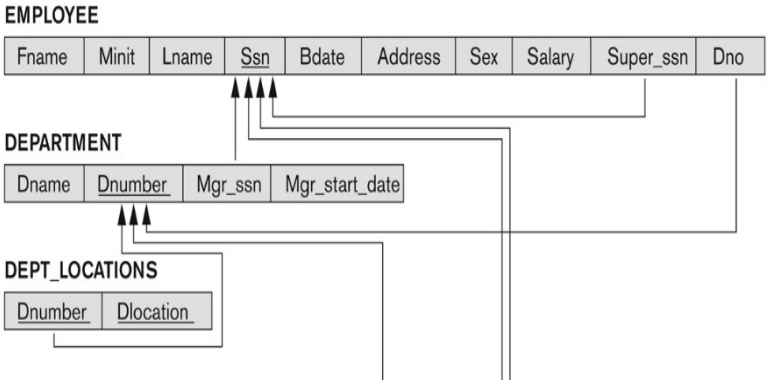


Mapping Regular Entity Types - Multi valued Attributes - Example

- The attribute **DLOCATION** represents the multi valued attribute LOCATIONS of DEPARTMENT.
- The relation **DEPT_LOCATIONS** is created.
- **DNUMBER** - foreign key-represents the primary key of the DEPARTMENT relation.
- The primary key of **DEPT_LOCATIONS** is the combination of DNUMBER, DLOCATION.

Mapping Regular Entity Types - Multivalued Attributes





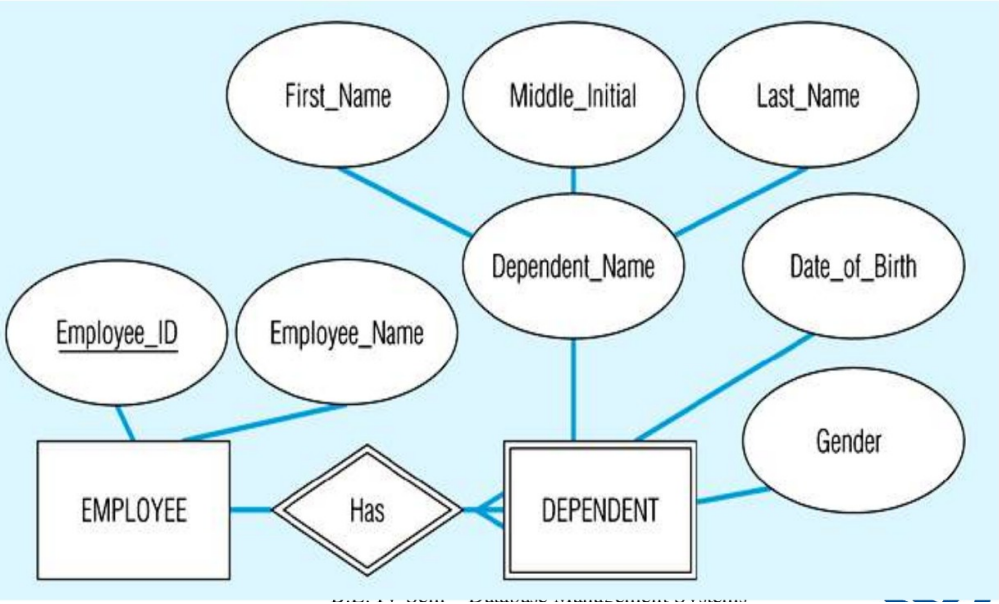
3. Mapping of Weak Entity Types

For each weak entity type W in the ER schema with owner entity type E:

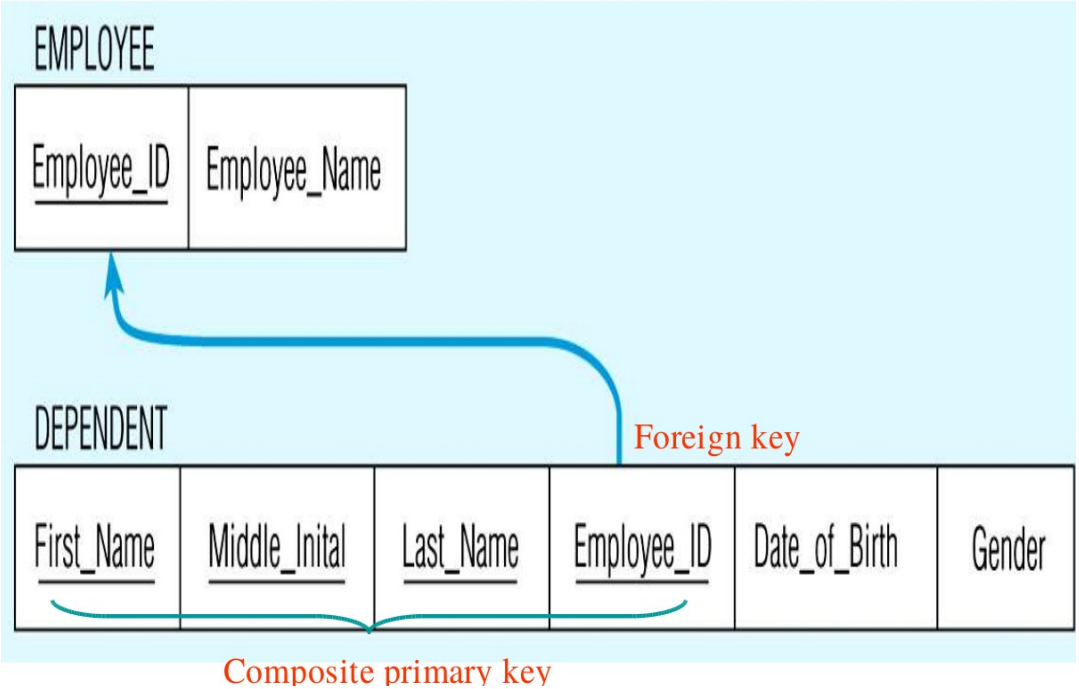
- Create a relation R and include all simple attributes (or simple components of composite attributes) of W as attributes of R.
- The primary key attribute(s) of the relation(s) correspond to the owner entity type(s), will be included as foreign key attributes of R.
- The primary key of R is the combination of the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.

Mapping of Weak Entity Types

Mapping weak entity types



Mapping of Weak Entity Types



Mapping of Weak Entity Types - Example

- Create the relation **DEPENDENT** which correspond to the weak entity type DEPENDENT.
- Include the primary key **SSN** of the EMPLOYEE relation as a **foreign key** attribute of DEPENDENT (renamed to ESSN).
- The primary key of the DEPENDENT relation is the combination **ESSN, DEPENDENT_NAME** because DEPENDENT_NAME is the partial key of DEPENDENT.

Mapping of Weak Entity Types

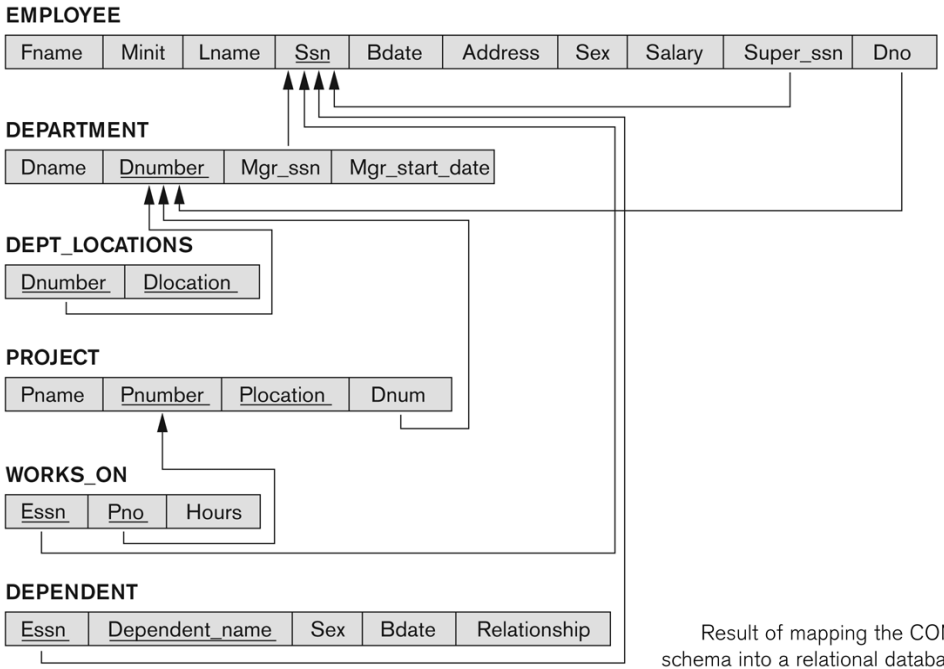


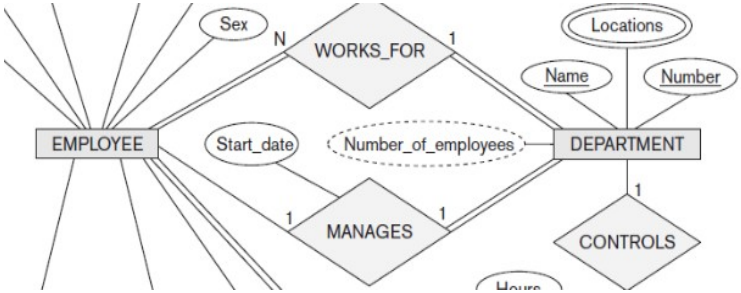
Figure 7.2
Result of mapping the COMPANY ER schema into a relational database schema.

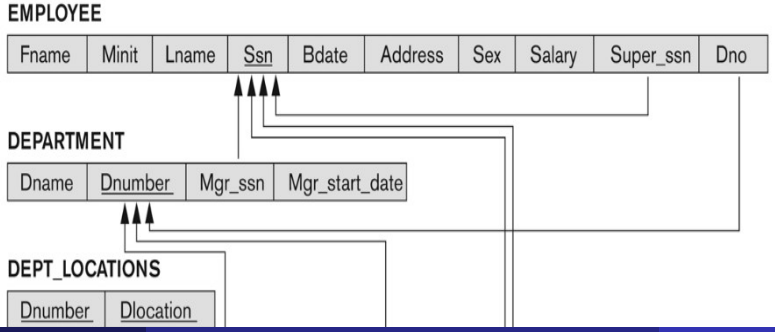
Mapping of Binary 1:1 Relation Types

- For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R.
  - **Foreign Key approach:** Choose one of the relations say S and include the primary key of T as a foreign key in S. Choose an entity type with total participation in R in the role of S.
  - **Merged relation option**: An alternate mapping of a 1:1 relationship type is by merging the two entity types and the relationship into a single relation. **(both participation's are total)**.
  - **Cross-reference or relationship relation option:** The third alternative is to set up a third relation R for the purpose of cross-referencing the primary keys of the two relations S and T representing the entity types. This approach suitable for binary M:N relationships.

4. Foreign Key approach - Example

Example: 1:1 relation MANAGES is mapped by choosing the entity type DEPARTMENT to serve in the role of S (total participation) mgr_ssn of S as fk
SSN in employee is pk.



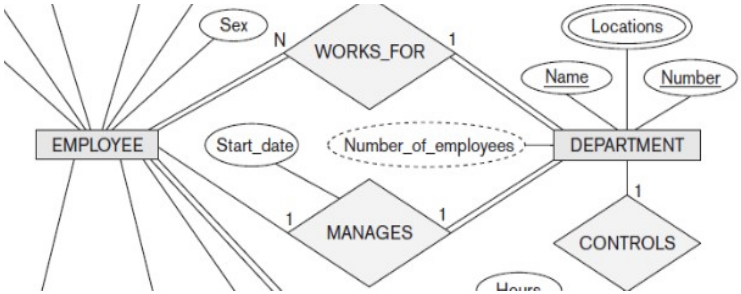


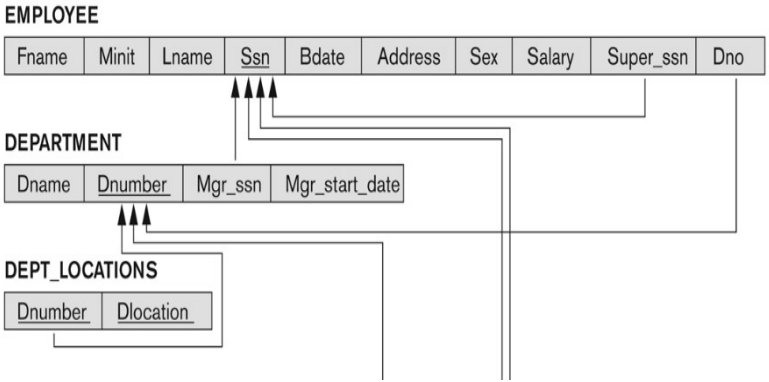
5. Mapping of Binary 1:N Relationship Types

For each regular binary 1:N relationship type R

- Identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include the primary key of the relation (1 side) T as as foreign key in S(N side).
- Include any simple attributes of the 1:N relation type as attributes of S.
- Example: 1:N relationship types **WORKS_FOR and CONTROLS,** .
- For **WORKS_FOR** include the primary key DNUMBER of the DEPARTMENT relation as foreign key in the EMPLOYEE relation as DNO.

Mapping of Binary 1:N Relationship Types





6. Mapping of Binary M:N Relationship Types

For each regular binary M:N relationship type R

- Create a new relation S to represent the realtionship type R.
- Include foreign key attributes in S as the primary keys of the relations that represent the participating entity types; their combination will form the primary key of S.
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S.

Step 6: Mapping of Binary M:N Relationship Types -Example

- The M:N relationship type **WORKS_ON** from the ER diagram is mapped by creating a relation WORKS_ON in the relational database schema.
- The primary keys of the PROJECT and EMPLOYEE relations are included as foreign keys in WORKS_ON and renamed **PNO and ESSN** respectively.
- Attribute HOURS in WORKS_ON represents the HOURS attribute of the relation type. The primary key of the WORKS_ON relation is the combination of the foreign key attributes ESSN, PNO.

Mapping of Binary M:N Relationship Types

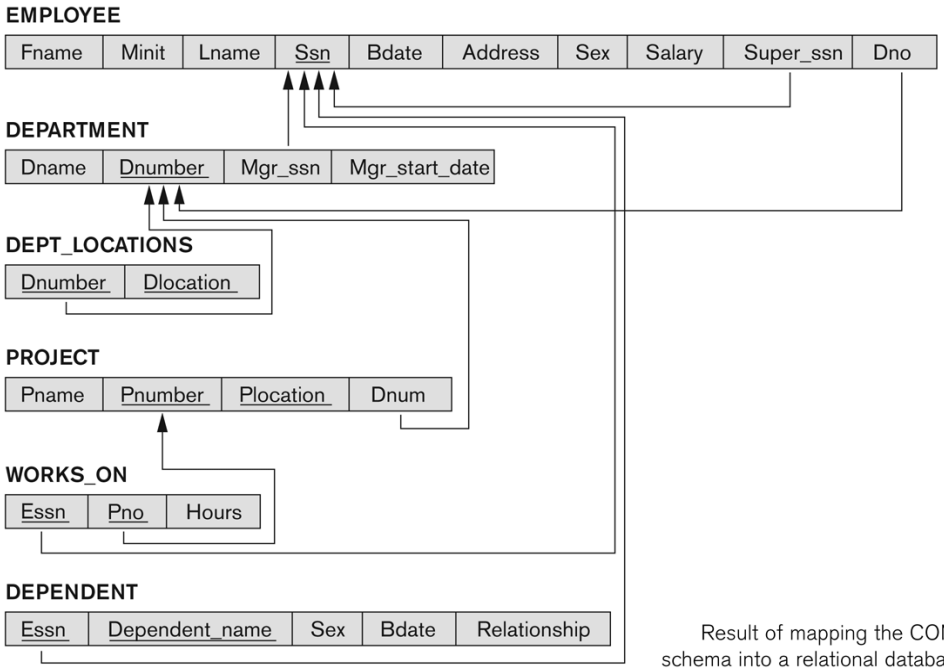


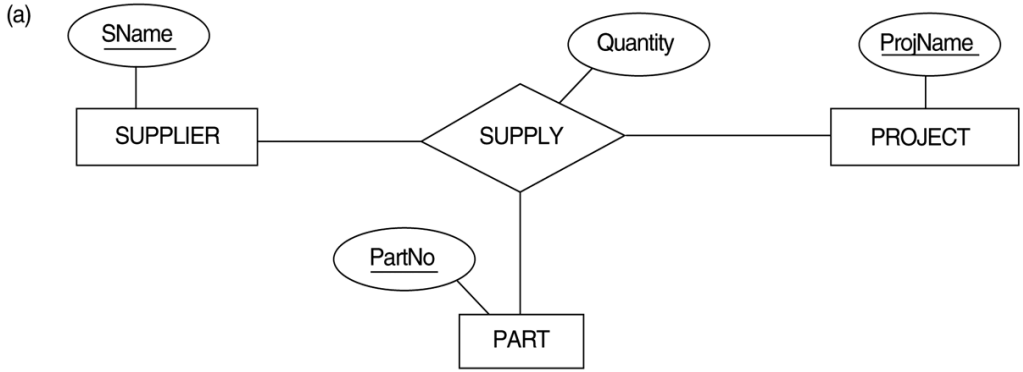
Figure 7.2
Result of mapping the COMPANY ER schema into a relational database schema.

7. Mapping of N-ary Relationship Types

For each n-ary relationship type R, where n>2,

- Create a new relationship S to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
- Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S. Example: The relationship type SUPPY in the ER.
- This can be mapped to the relation SUPPLY shown in the relational schema, whose primary key is the combination of the three foreign keys SNAME, PARTNO, PROJNAME

Mapping of N-ary Relationship Types



Mapping of N-ary Relationship Types

SUPPLIER

SNAME	. . .

PROJECT

PROJNAME	. . .

PART

PARTNO	. . .

SUPPLY

SNAME	PROJNAME	PARTNO	QUANTITY

Correspondence between ER and Relational Models

ER Model	Relational Model
Entity type	"Entity" relation
1:1 or 1:N relationship type	Foreign key (or "relationship" relation)
M:N relationship type	"Relationship" relation and two foreign keys
n-ary relationship type	"Relationship" relation and n foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key

Reference

Fundamentals of Database systems 7th Edition by Ramez Elmasri.